

Candy Mini-Project

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```
candy <- read.csv("C:/Users/chenr/Downloads/candy-data.csv", row.names = 1)
head(candy)
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer
## 100 Grand           1      0           1              0      0              1
## 3 Musketeers        1      0           0              0      1              0
## One dime            0      0           0              0      0              0
## One quarter         0      0           0              0      0              0
## Air Heads           0      1           0              0      0              0
## Almond Joy          1      0           0              1      0              0
##           hard bar pluribus sugarpercent pricepercent winpercent
## 100 Grand      0  1           0          0.732      0.860    66.97173
## 3 Musketeers   0  1           0          0.604      0.511    67.60294
## One dime       0  0           0          0.011      0.116    32.26109
## One quarter    0  0           0          0.011      0.511    46.11650
## Air Heads      0  0           0          0.906      0.511    52.34146
## Almond Joy     0  1           0          0.465      0.767    50.34755
```

```
library(flextable)
flextable::flextable(head(candy, 10))
```

chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard	bar	pluribus	sug
1	0	1	0	0	1	0	1	0	
1	0	0	0	1	0	0	1	0	
0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	
0	1	0	0	0	0	0	0	0	
1	0	0	1	0	0	0	1	0	
1	0	1	1	1	0	0	1	0	
0	0	0	1	0	0	0	0	1	
0	0	0	0	0	0	0	0	1	
0	1	1	0	0	0	0	0	0	

What is in the dataset?

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
## [1] 85
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.2      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x purrr::compose() masks flextable::compose()
```

```
## x dplyr::filter()  masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
candy %>%
```

```
  nrow()
```

```
## [1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
## [1] 38
```

2.2 What is your favorite candy?

Q3. What is your favorite candy in the dataset and what is its winpercent value?

My favorite winpercent

```
candy["Hershey's Milk Chocolate", ]$winpercent
```

```
## [1] 56.4905
```

```
library(dplyr)
```

```
candy |>
```

```
  filter(rownames(candy) == "Hershey's Milk Chocolate") |>
```

```
  select(winpercent)
```

```
##               winpercent
```

```
## Hershey's Milk Chocolate 56.4905
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy |>
```

```
  filter(rownames(candy) == "Kit Kat") |>
```

```
  select(winpercent)
```

```
##               winpercent
```

```
## Kit Kat         76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy |>
```

```
  filter(rownames(candy) == "Tootsie Roll Snack Bars") |>
```

```
  select(winpercent)
```

```
##                                winpercent
## Tootsie Roll Snack Bars      49.6535
```

Side-note: the `skimr::skim()` function

```
library(skimr)
skimr::skim(candy)
```

Table 2: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	
	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent is on 0-100 scale and the rest are 0-1 scale.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

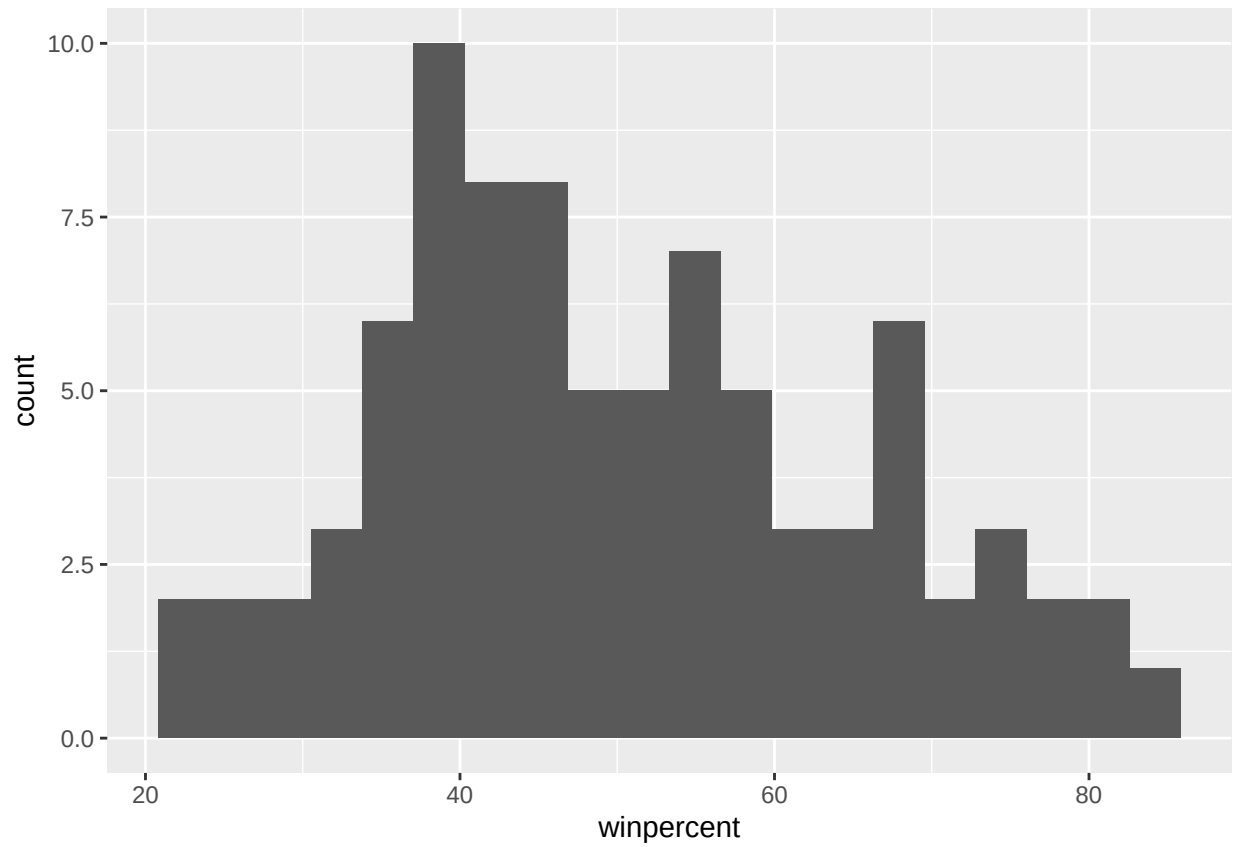
That the candy does not contain chocolate.

3.Exploratory analysis

Q8. Plot a histogram of winpercent values

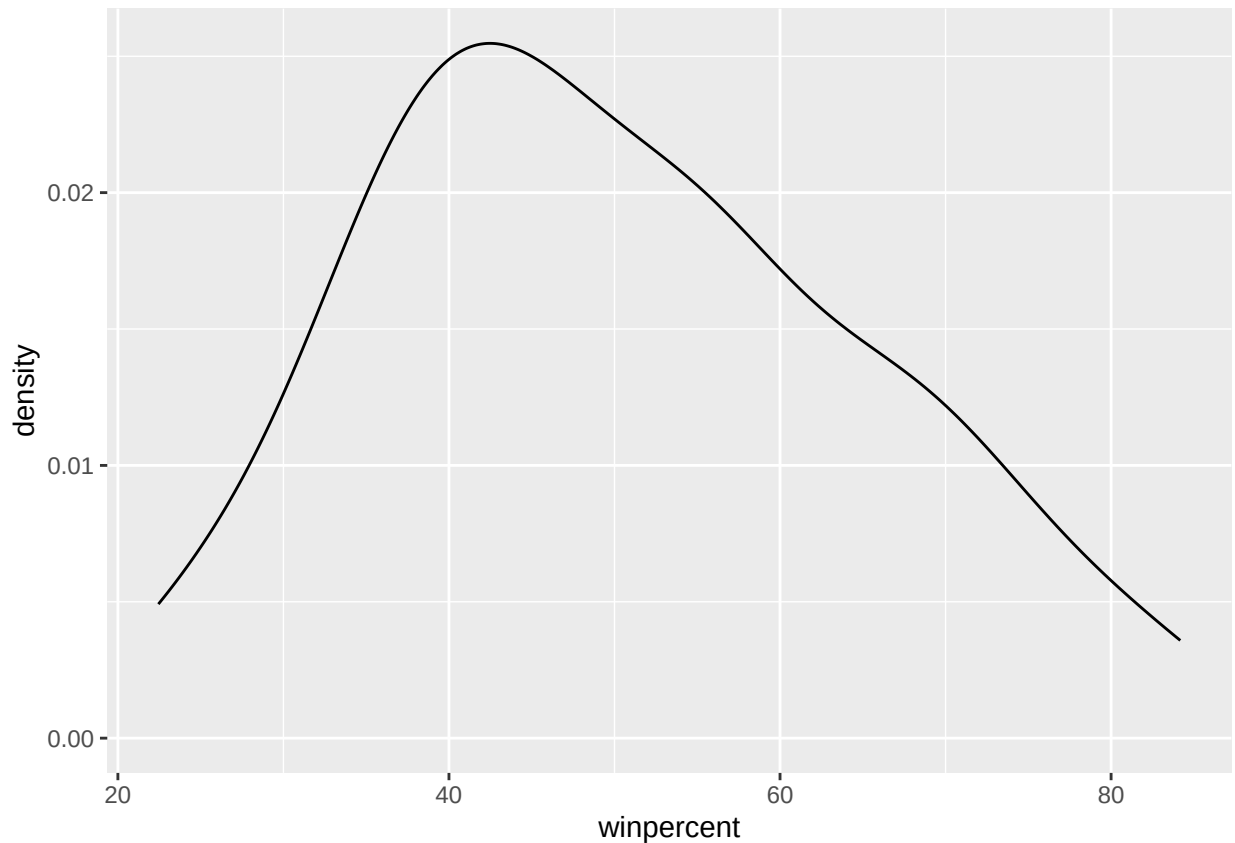
```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=20)
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_density()
```



Q9. Is the distribution of winpercent values symmetrical?

Based on the summary statistics, the mean (50.32) exceeds the median (47.83), and the maximum value (84.18) lies much farther above the median than the minimum (22.45) lies below it. This indicates that the distribution is not symmetric and is positively (right) skewed.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
## [1] 50.31676
```

```
summary(candy$winpercent)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  22.45   39.14   47.83   50.32   59.86   84.18
```

The median winpercent is 47.83, which is below 50%. Therefore, the center of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Chocolate is higher ranked on average.

```
choc.inds <- candy$chocolate==1
choc.win  <- candy[choc.inds,]$winpercent
choc.win  <- mean(choc.win)
choc.win
```

```
## [1] 60.92153
```

```
mean(candy[candy$chocolate==1,]$winpercent)
```

```
## [1] 60.92153
```

```
mean(candy[candy$fruity==1,]$winpercent)
```

```
## [1] 44.11974
```

```
fruit.ind <- candy$fruity==1  
fruit.win <- candy[fruit.ind,]$winpercent  
fruit.mean <- mean(fruit.win)
```

```
choc.ind <- candy$chocolate==1  
choc.win <- candy[choc.ind,]$winpercent  
choc.mean <- mean(choc.win)
```

```
candy |>  
  filter(chocolate == 1) |>  
  select(winpercent)
```

##	winpercent
## 100 Grand	66.97173
## 3 Musketeers	67.60294
## Almond Joy	50.34755
## Baby Ruth	56.91455
## Charleston Chew	38.97504
## Hershey's Kisses	55.37545
## Hershey's Krackel	62.28448
## Hershey's Milk Chocolate	56.49050
## Hershey's Special Dark	59.23612
## Junior Mints	57.21925
## Kit Kat	76.76860
## Peanut butter M&M's	71.46505
## M&M's	66.57458
## Milk Duds	55.06407
## Milky Way	73.09956
## Milky Way Midnight	60.80070
## Milky Way Simply Caramel	64.35334
## Mounds	47.82975
## Mr Good Bar	54.52645
## Nestle Butterfinger	70.73564
## Nestle Crunch	66.47068
## Peanut M&Ms	69.48379
## Reese's Miniatures	81.86626
## Reese's Peanut Butter cup	84.18029
## Reese's pieces	73.43499
## Reese's stuffed with pieces	72.88790
## Rolo	65.71629
## Sixlets	34.72200
## Nestle Smarties	37.88719
## Snickers	76.67378
## Snickers Crisper	59.52925
## Tootsie Pop	48.98265
## Tootsie Roll Juniors	43.06890
## Tootsie Roll Midgies	45.73675

```
## Tootsie Roll Snack Bars      49.65350
## Twix                        81.64291
## Whoppers                    49.52411
```

Q12. Is this difference statistically significant?

Yes, the difference is statistically significant.

```
t.test(choc.win, fruit.win)
```

```
##
## Welch Two Sample t-test
##
## data: choc.win and fruit.win
## t = 6.2582, df = 68.882, p-value = 2.871e-08
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  11.44563 22.15795
## sample estimates:
## mean of x mean of y
##  60.92153 44.11974
```

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  head(5)
```

```
##               chocolate fruity caramel peanutyalmondy nougat
## Nik L Nip           0      1      0                0      0
## Boston Baked Beans  0      0      0                1      0
## Chiclets           0      1      0                0      0
## Super Bubble       0      1      0                0      0
## Jawbusters         0      1      0                0      0
##
##      crispedricewafer hard bar pluribus sugarpercent pricepercent
## Nik L Nip                0      0      0      1      0.197      0.976
## Boston Baked Beans      0      0      0      1      0.313      0.511
## Chiclets                0      0      0      1      0.046      0.325
## Super Bubble            0      0      0      0      0.162      0.116
## Jawbusters              0      1      0      1      0.093      0.511
##
##               winpercent
## Nik L Nip      22.44534
## Boston Baked Beans 23.41782
## Chiclets      24.52499
## Super Bubble   27.30386
## Jawbusters     28.12744
```

```
x <- c(5,1,10,4)
sort(x)
```

```
## [1]  1  4  5 10
```

```
ord.ind <- order(candy$winpercent)
head(candy[ord.ind,],5)
```

```
##               chocolate fruity caramel peanutyalmondy nougat
```

```
## Nik L Nip          0      1      0          0      0
## Boston Baked Beans 0      0      0          1      0
## Chiclets          0      1      0          0      0
## Super Bubble      0      1      0          0      0
## Jawbusters        0      1      0          0      0
##
##      crispedricewafer hard bar pluribus sugarpercent pricepercent
## Nik L Nip          0      0      0          1      0.197      0.976
## Boston Baked Beans 0      0      0          1      0.313      0.511
## Chiclets          0      0      0          1      0.046      0.325
## Super Bubble      0      0      0          0      0.162      0.116
## Jawbusters        0      1      0          1      0.093      0.511
##
##      winpercent
## Nik L Nip      22.44534
## Boston Baked Beans 23.41782
## Chiclets      24.52499
## Super Bubble   27.30386
## Jawbusters     28.12744
```

#five least favorite candy

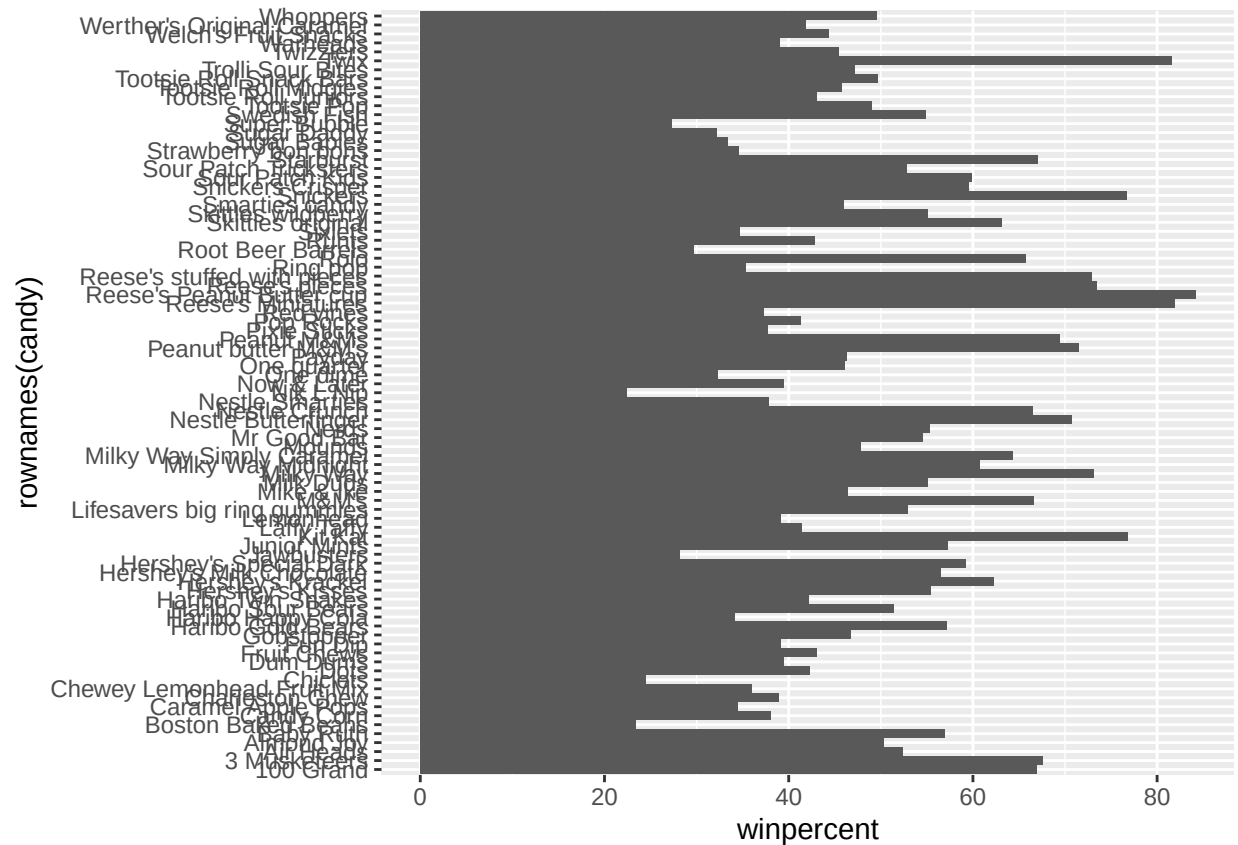
Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(-winpercent) |>
  head(5)
```

```
##
##      chocolate fruity caramel peanutyalmondy nougat
## Reese's Peanut Butter cup      1      0      0          1      0
## Reese's Miniatures             1      0      0          1      0
## Twix                           1      0      1          0      0
## Kit Kat                       1      0      0          0      0
## Snickers                      1      0      1          1      1
##
##      crispedricewafer hard bar pluribus sugarpercent
## Reese's Peanut Butter cup      0      0      0          0      0.720
## Reese's Miniatures             0      0      0          0      0.034
## Twix                           1      0      1          0      0.546
## Kit Kat                       1      0      1          0      0.313
## Snickers                      0      0      1          0      0.546
##
##      pricepercent winpercent
## Reese's Peanut Butter cup      0.651  84.18029
## Reese's Miniatures             0.279  81.86626
## Twix                           0.906  81.64291
## Kit Kat                       0.511  76.76860
## Snickers                      0.651  76.67378
```

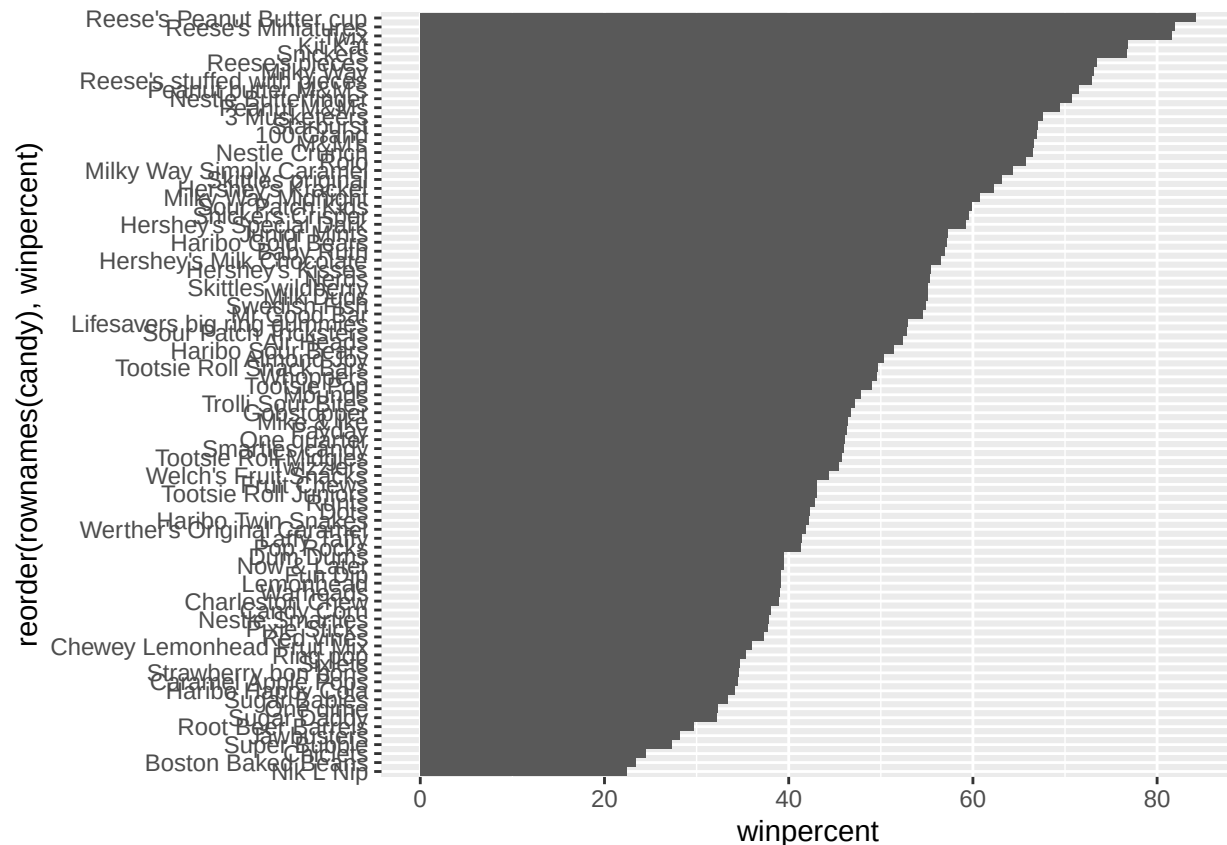
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +
  aes(x=winpercent,
      y=reorder(rownames(candy), winpercent)) +
  geom_col()
```



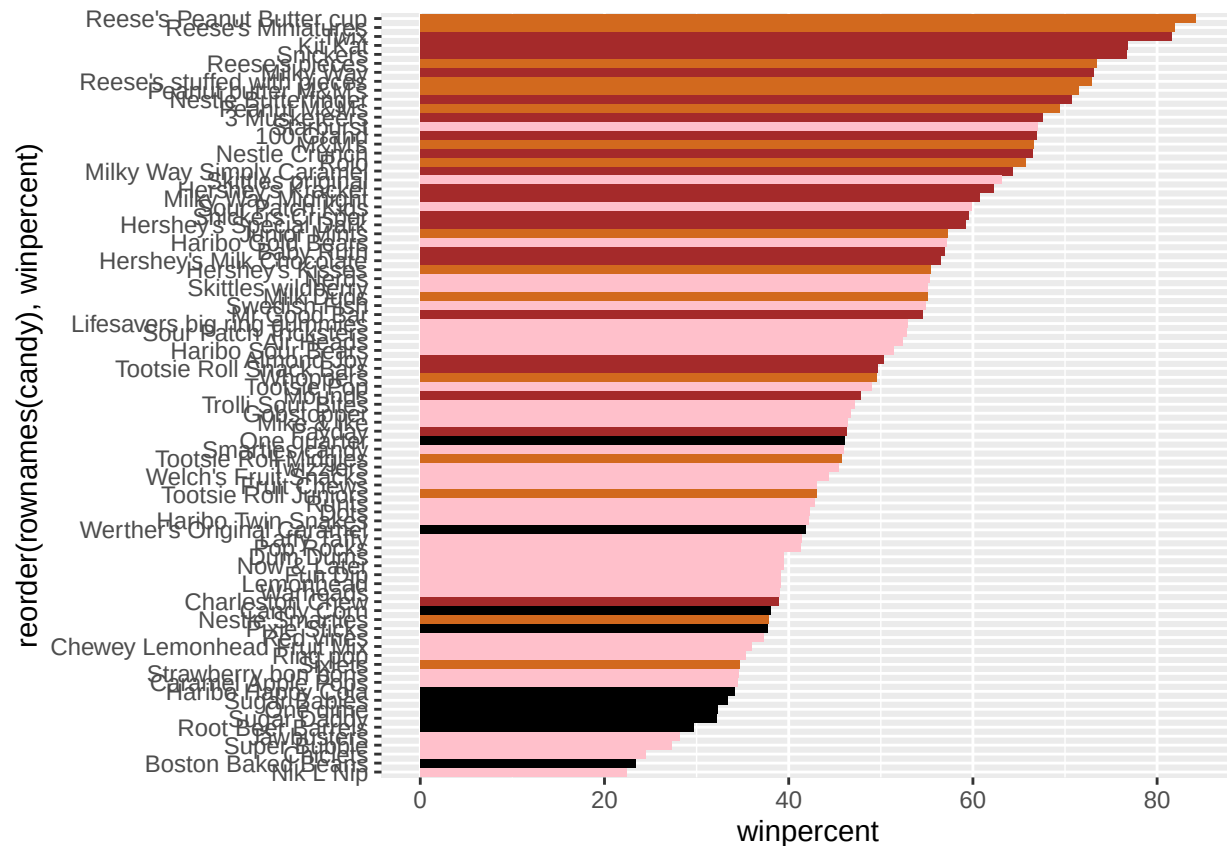
4.0.1 Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$fruity)] = "pink"
my_cols[as.logical(candy$bar)] = "brown"
my_cols
```

```
## [1] "brown" "brown" "black" "black" "pink" "brown"
## [7] "brown" "black" "black" "pink" "brown" "pink"
## [13] "pink" "pink" "pink" "pink" "pink" "pink"
## [19] "pink" "black" "pink" "pink" "chocolate" "brown"
## [25] "brown" "brown" "pink" "chocolate" "brown" "pink"
## [31] "pink" "pink" "chocolate" "chocolate" "pink" "chocolate"
## [37] "brown" "brown" "brown" "brown" "brown" "pink"
## [43] "brown" "brown" "pink" "pink" "brown" "chocolate"
## [49] "black" "pink" "pink" "chocolate" "chocolate" "chocolate"
## [55] "chocolate" "pink" "chocolate" "black" "pink" "chocolate"
## [61] "pink" "pink" "chocolate" "pink" "brown" "brown"
## [67] "pink" "pink" "pink" "pink" "black" "black"
## [73] "pink" "pink" "pink" "chocolate" "chocolate" "brown"
## [79] "pink" "brown" "pink" "pink" "pink" "black"
## [85] "chocolate"
```

```
ggplot(candy) +
  aes(x=winpercent,
```

```
y=reorder(rownames(candy), winpercent)) +  
geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

```
candy |>  
  dplyr::filter(chocolate == 1) |>  
  dplyr::slice_min(winpercent, n = 1, with_ties = FALSE)
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard  
## Sixlets           1         0         0         0         0         0         0  
##           bar pluribus sugarpercent pricepercent winpercent  
## Sixlets           0         1         0.22         0.081         34.722
```

Q18. What is the best ranked fruity candy?

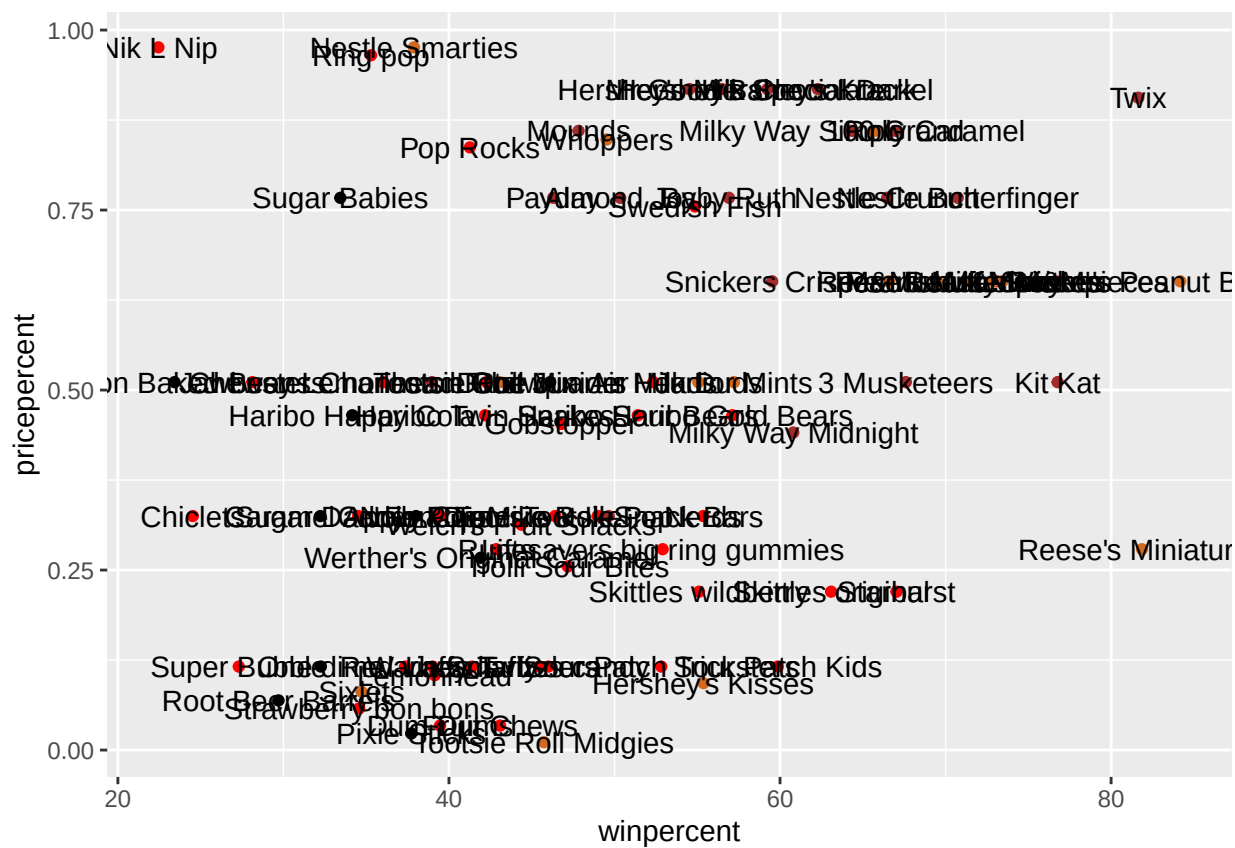
```
candy |>  
  dplyr::filter(fruity == 1) |>  
  dplyr::slice_max(winpercent, n = 1, with_ties = FALSE)
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard  
## Starburst           0         1         0         0         0         0         0  
##           bar pluribus sugarpercent pricepercent winpercent  
## Starburst           0         1         0.151         0.22         67.03763
```

5. Taking a look at pricepercent

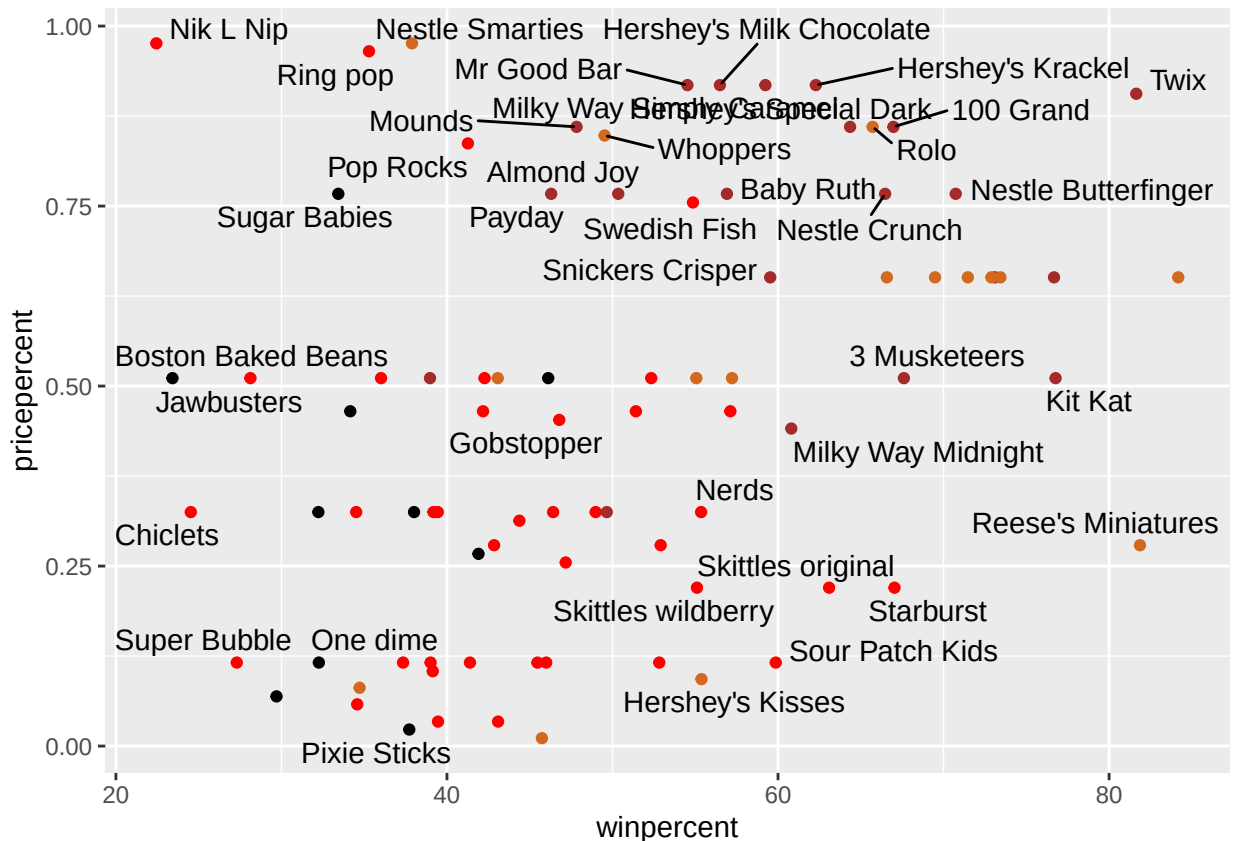
```
my_cols[as.logical(candy$fruity)] = "red"
```

```
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=row.names(candy)) +
  geom_point(col=my_cols) +
  geom_text()
```



```
library(ggrepel)
```

```
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=row.names(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(max.overlaps = 7)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
candy |>
  dplyr::mutate(type = dplyr::case_when(chocolate == 1 & bar == 1 ~ "chocolate bar",
                                         chocolate == 1 & bar == 0 ~ "chocolate other",
                                         fruity == 1 ~ "fruity",
                                         TRUE ~ "other")) |>

  dplyr::group_by(type) |>
  dplyr::summarise(mean_win = mean(winpercent, na.rm = TRUE),
                   mean_price = mean(pricepercent, na.rm = TRUE), .groups = "drop") |>
  dplyr::mutate(score = dplyr::min_rank(mean_price) + dplyr::min_rank(dplyr::desc(mean_win))) |>
  dplyr::slice_min(score, n = 1, with_ties = FALSE)
```

```
## # A tibble: 1 x 4
##   type    mean_win mean_price score
##   <chr>    <dbl>    <dbl> <int>
## 1 fruity    44.0      0.333     4
```

Fruity type is the highest ranked in terms of winpercent for the least money.

```
candy |>
  dplyr::mutate(score = dplyr::min_rank(pricepercent) +
                 dplyr::min_rank(dplyr::desc(winpercent))) |>
  dplyr::slice_min(score, n = 1, with_ties = FALSE)
```

```
##               chocolate fruity caramel peanutyalmondy nougat
## Reese's Miniatures           1         0         0           1         0
```

```
##               crispedricewafer hard bar pluribus sugarpercent pricepercent
## Reese's Miniatures           0   0   0           0         0.034         0.279
##               winpercent score
## Reese's Miniatures    81.86626    26
```

Reese's Miniatures is the highest single candy ranked in terms of winpercent for the least money

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy |>
  arrange(-pricepercent) |>
  head(5)
```

```
##               chocolate fruity caramel peanutyalmondy nougat
## Nik L Nip           0       1         0                0       0
## Nestle Smarties     1       0         0                0       0
## Ring pop            0       1         0                0       0
## Hershey's Krackel   1       0         0                0       0
## Hershey's Milk Chocolate 1       0         0                0       0
##               crispedricewafer hard bar pluribus sugarpercent
## Nik L Nip                0   0   0           1         0.197
## Nestle Smarties          0   0   0           1         0.267
## Ring pop                 0   1   0           0         0.732
## Hershey's Krackel        1   0   1           0         0.430
## Hershey's Milk Chocolate 0   0   1           0         0.430
##               pricepercent winpercent
## Nik L Nip           0.976    22.44534
## Nestle Smarties     0.976    37.88719
## Ring pop            0.965    35.29076
## Hershey's Krackel   0.918    62.28448
## Hershey's Milk Chocolate 0.918    56.49050
```

```
candy |>
  arrange(-pricepercent) |>
  head(5) |>
  dplyr::slice_min(winpercent, n = 1, with_ties = FALSE)
```

```
##               chocolate fruity caramel peanutyalmondy nougat crispedricewafer hard
## Nik L Nip           0       1         0                0       0
##               bar pluribus sugarpercent pricepercent winpercent
## Nik L Nip    0           1         0.197         0.976    22.44534
```

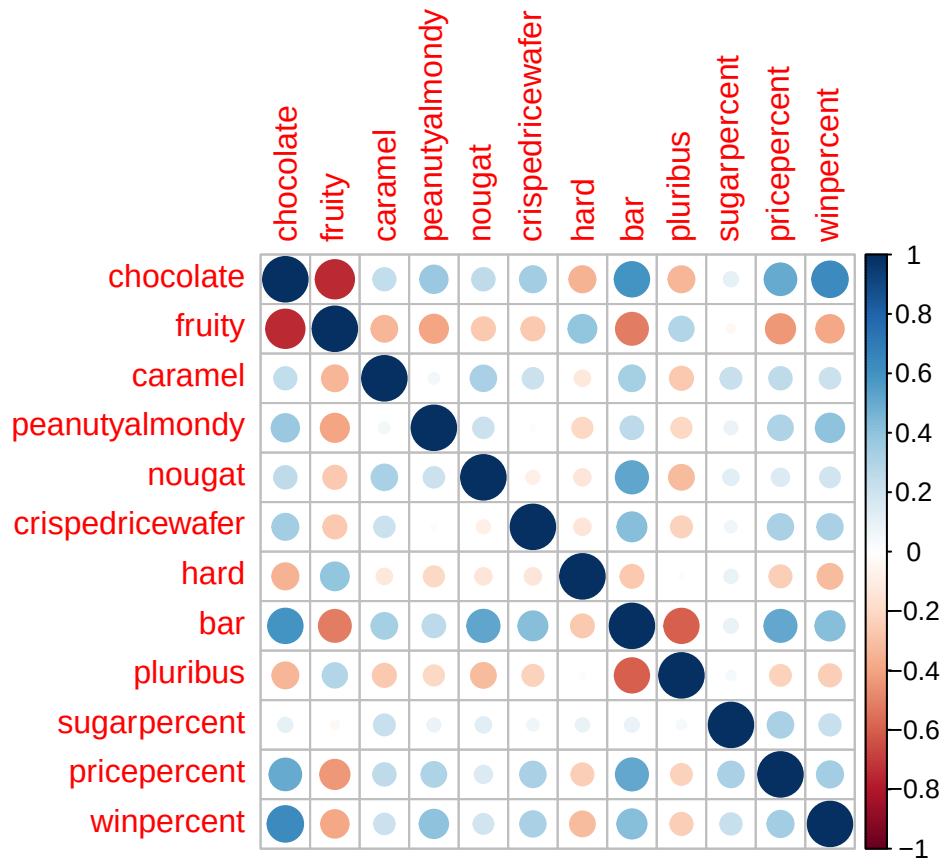
Nik L Nip, Ring Pop, Nestlé Smarties, Hershey's Krackel, Hershey's Milk Chocolate. Among these, the least popular (lowest winpercent) is Nik L Nip.

6. Exploring the correlation structure

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



cij

```
##          chocolate    fruity    caramel    peanutyalmondy    nougat
## chocolate    1.0000000 -0.74172106  0.24987535    0.37782357  0.25489183
## fruity       -0.7417211  1.00000000 -0.33548538   -0.39928014 -0.26936712
## caramel      0.2498753 -0.33548538  1.00000000    0.05935614  0.32849280
## peanutyalmondy 0.3778236 -0.39928014  0.05935614    1.00000000  0.21311310
## nougat       0.2548918 -0.26936712  0.32849280    0.21311310  1.00000000
## crispedricewafer 0.3412098 -0.26936712  0.21311310   -0.01764631 -0.08974359
## hard        -0.3441769  0.39067750 -0.12235513   -0.20555661 -0.13867505
## bar         0.5974211 -0.51506558  0.33396002    0.26041960  0.52297636
## pluribus     -0.3396752  0.29972522 -0.26958501   -0.20610932 -0.31033884
## sugarpercent  0.1041691 -0.03439296  0.22193335    0.08788927  0.12308135
## pricepercent  0.5046754 -0.43096853  0.25432709    0.30915323  0.15319643
## winpercent    0.6365167 -0.38093814  0.21341630    0.40619220  0.19937530
##          crispedricewafer    hard    bar    pluribus
## chocolate    0.34120978 -0.34417691  0.59742114 -0.33967519
## fruity       -0.26936712  0.39067750 -0.51506558  0.29972522
## caramel      0.21311310 -0.12235513  0.33396002 -0.26958501
## peanutyalmondy -0.01764631 -0.20555661  0.26041960 -0.20610932
## nougat       -0.08974359 -0.13867505  0.52297636 -0.31033884
## crispedricewafer 1.00000000 -0.13867505  0.42375093 -0.22469338
## hard        -0.13867505  1.00000000 -0.26516504  0.01453172
## bar         0.42375093 -0.26516504  1.00000000 -0.59340892
## pluribus     -0.22469338  0.01453172 -0.59340892  1.00000000
## sugarpercent  0.06994969  0.09180975  0.09998516  0.04552282
```

```
## pricepercent      0.32826539 -0.24436534  0.51840654 -0.22079363
## winpercent       0.32467965 -0.31038158  0.42992933 -0.24744787
##               sugarpercent pricepercent winpercent
## chocolate      0.10416906   0.5046754  0.6365167
## fruity         -0.03439296  -0.4309685 -0.3809381
## caramel        0.22193335   0.2543271  0.2134163
## peanutyalmondy  0.08788927   0.3091532  0.4061922
## nougat         0.12308135   0.1531964  0.1993753
## crispedricewafer 0.06994969   0.3282654  0.3246797
## hard           0.09180975  -0.2443653 -0.3103816
## bar            0.09998516   0.5184065  0.4299293
## pluribus       0.04552282  -0.2207936 -0.2474479
## sugarpercent    1.00000000   0.3297064  0.2291507
## pricepercent    0.32970639   1.0000000  0.3453254
## winpercent      0.22915066   0.3453254  1.0000000
```

Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolaty and Fruity, bar and pluribus. etc.

Q23. Similarly, what two variables are most positively correlated?

Chocolaty and bar, fruity and hard, etc.

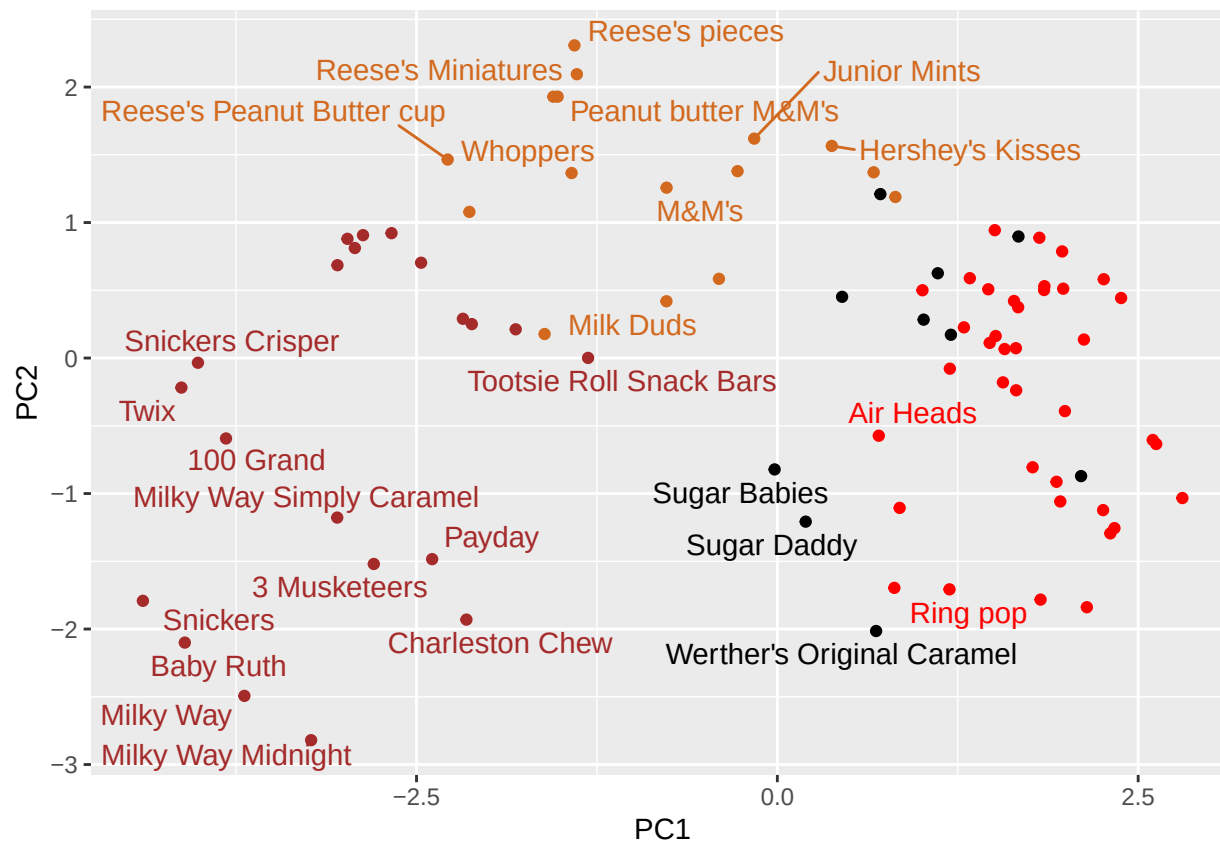
7. Principle Component Analysis

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

```
## Importance of components:
##               PC1    PC2    PC3    PC4    PC5    PC6    PC7
## Standard deviation  2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
## Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
## Cumulative Proportion 0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
##               PC8    PC9    PC10    PC11    PC12
## Standard deviation  0.74530 0.67824 0.62349 0.43974 0.39760
## Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
## Cumulative Proportion 0.89998 0.93832 0.97071 0.98683 1.00000
```

Main PCA result figure

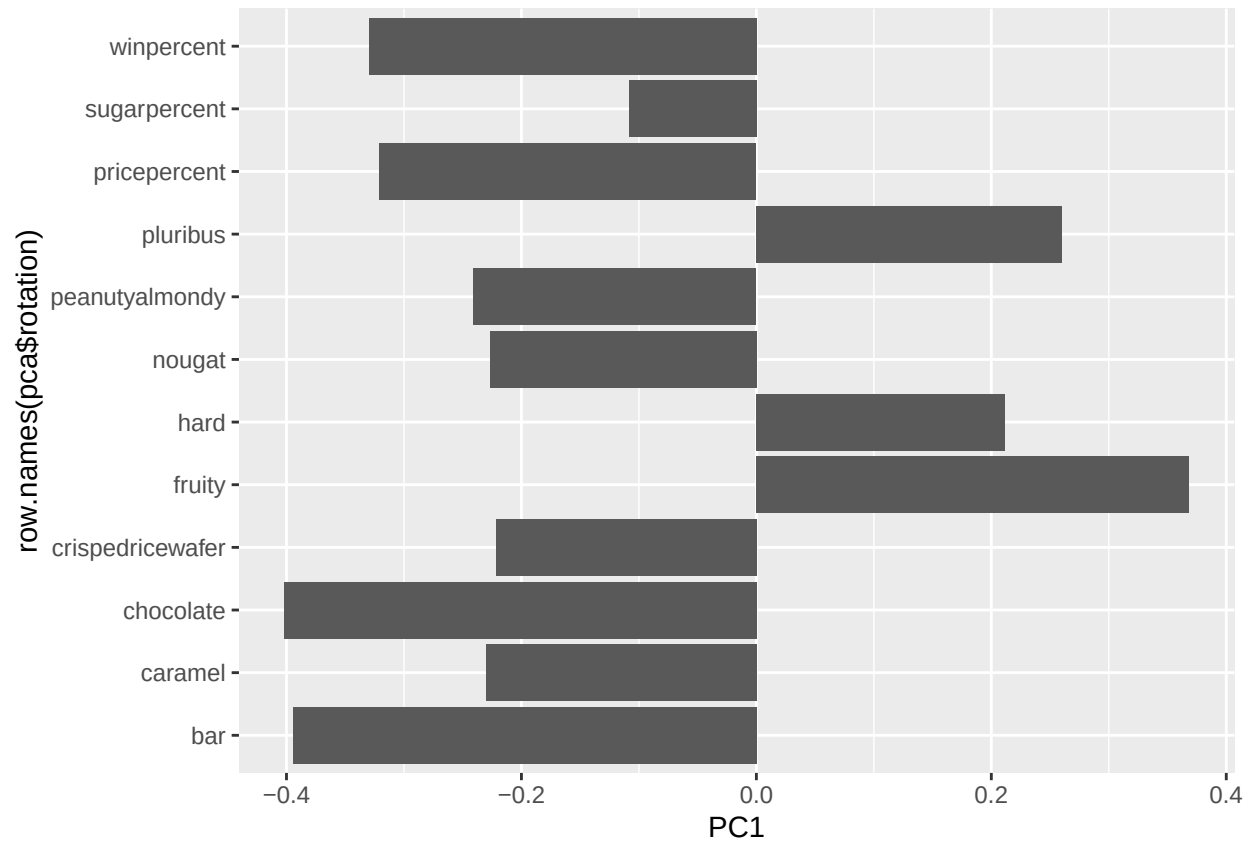
```
ggplot(pca$x, aes(PC1, PC2)) +
  geom_point(col=my_cols) +
  geom_text_repel(aes(label = rownames(pca$x)), col = my_cols, max.overlaps = 5)
```

```
library(plotly)
```

```
##
## Attaching package: 'plotly'
## The following object is masked from 'package:ggplot2':
##
##   last_plot
## The following objects are masked from 'package:flextable':
##
##   highlight, style
## The following object is masked from 'package:stats':
##
##   filter
## The following object is masked from 'package:graphics':
##
##   layout
```

```
ggplot(pca$rotation)+
  aes(PC1, row.names(pca$rotation))+
  geom_col()
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

Fruity, pluribus, and hard load strongly positively on PC1.

PC1 distinguishes non-chocolate, fruit-flavored, hard, multipack candies (positive side) from chocolate, bar-type, higher price, and higher win-percent features (negative side), reflecting the known contrast between fruity and chocolate candies and the similarity between chocolate and bar characteristics.